Week 10 – POWER BI TASK

Real Estate Analytics Dashboard

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Objective

The objective of this project is to analyze house sales data using Power BI. The dashboard helps understand property prices, compare house features, and explore property prices across different locations.

Data Preparation

The dataset was imported into Power BI and prepared for visualization.

The following tasks were performed:

- Checked the column names and data types.
- Prepared the required numerical fields.
- Used the existing calculated columns and fields for analysis.
- Created DAX measures to calculate important property statistics.

DAX Measures Created

The following measures were created:

1. Total Properties Sold – Counts the total number of properties.
2. Average House Price – Calculates the average property price.
3. Median Price – Calculates the median property price.
4. Average Living Area – Calculates the average living area.
5. Average Price per SqFt – Calculates the average price per square foot.

Dashboard Visualizations

The following visualizations were created:

KPI Cards

Five KPI cards were created to display important information:

- Total Properties Sold
- Average House Price
- Median Price
- Average Living Area

- Average Price per SqFt

Property Prices by Location:

A map visual was created using latitude and longitude to display property locations. Price Tier was used as the legend to distinguish different property price categories.

Average House Price by Bedrooms:

A clustered column chart was created to compare the average house price based on the number of bedrooms.

Average House Price by Bathrooms:

A clustered column chart was created to compare average house prices based on the number of bathrooms.

Average House Price by Waterfront Status:

A clustered column chart was created to compare the average prices of waterfront and non-waterfront properties.

Average House Price by Grade:

A clustered column chart was created to analyze the relationship between house grade and average property price.

Average House Price by Condition:

A clustered bar chart was created to compare average property prices across different house condition ratings.

Living Area Filter:

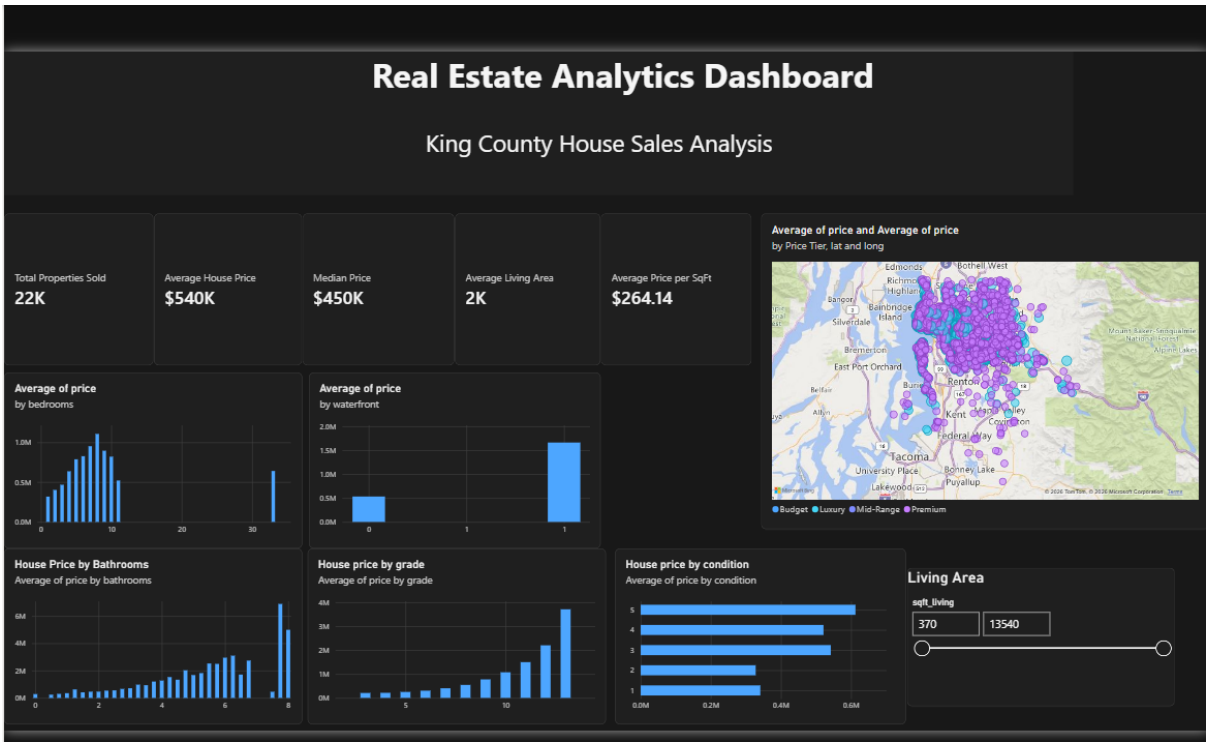
A numeric range slicer was added using the sqft_living field. It allows users to filter properties based on their living area.

Key Insights:

- The dashboard displays approximately 22,000 properties.
- The average house price is approximately $540K.
- The median house price is approximately $450K.
- The average living area is approximately 2,000 square feet.
- The average price per square foot is approximately $264.14.
- Property prices vary across different bedroom and bathroom counts.
- Waterfront status and house grade provide useful ways to compare property prices.
- The map helps identify the geographical distribution of properties.

Final Dashboard

All KPI cards, charts, the geographic map, and the living area slicer were arranged on a single interactive dashboard page.



Conclusion

The Real Estate Analytics Dashboard was successfully developed using Microsoft Power BI. It provides a clear overview of house sales data through KPI cards, charts, a geographical map, and an interactive slicer. This dashboard makes it easier to explore property prices and compare different housing features.